

meaning of truth changes between nodes. Further optimisation is also required for updating local alphas, as (87) may contain an embedded alpha on the left-hand side of the inequality — i.e. representative of all sub-formulae, which differs from the formula/operator alpha on the right-hand side. To achieve this, an alpha per node potentially requires a two-stage update of weights and alpha's independently of one another while enforcing constraints — future work.

There is an additional motivation for including an alpha per node that needs to be considered, namely, the training of an LNN with sub-symbolic/real-valued inputs. Under this configuration, a neural network may offer real-valued truth value bounds as facts, with LNN rules typically initialised as `True`. While the truth of these formulae are ordinarily used to self-supervise the LNN and minimise contradictions, they may also provide classical supervised target bounds to converge towards. With static bounds, weights parameters alone cannot change the truth of facts themselves. It must also be noted that while facts may be directly modifiable in the LNN framework, not all facts should be modifiable and the database designer may choose not to do so — e.g. given trusted fact bounds or operating the LNN in a regulated context. A node per alpha can therefore facilitate such a correction, i.e. turning the secondary fuzzy bounds in table 3 into classical bounds by decreasing α . Additional loss terms may also be required to ensure a balanced alpha per node, i.e. lowering α to increase the learning rate and allowing the defuzzification of bounds vs. increasing α to decrease the relative interpretability of truth between neurons.

F.6 Activation function independence

The tailored activation approach is defined using four boundary points. This permits logical operations to be expressed as any linear or non-linear function that intersects these points for upward inference.

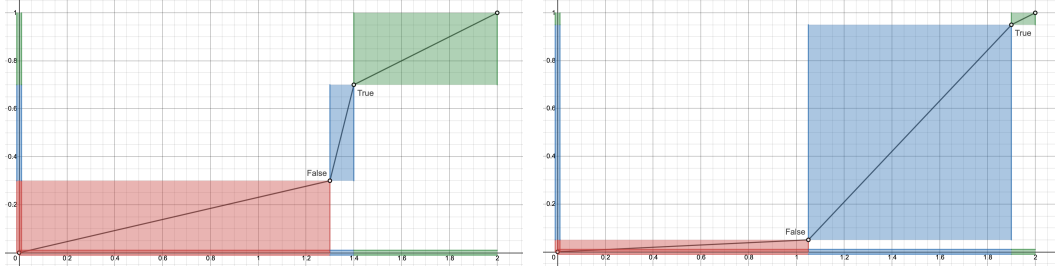


Figure 3: Piecewise linear activation function; left: $\alpha = 0.7$, right: $\alpha = 0.95$

The unique piece-wise linear function and functional inverse, is defined as:

$$\begin{aligned}
 (x_{min}, y_{min}) &= (0, 0) \\
 (x_F, y_F) &= \left(\sum_{i \in I} w_i - \alpha w_{max}, 1 - \alpha \right) \\
 (x_T, y_T) &= \left(\sum_{i \in I} w_i \cdot \alpha, \alpha \right) \\
 (x_{max}, y_{max}) &= \left(\sum_{i \in I} w_i, 1 \right) \\
 m_F &= \frac{y_F}{x_F} = \frac{1 - \alpha}{\sum_{i \in I} w_i - \alpha w_{max}} \\
 m_Z &= \frac{y_T - y_F}{x_T - x_F} = \frac{2\alpha - 1}{\alpha w_{max} - \sum_{i \in I} w_i (1 - \alpha)} \\
 m_T &= \frac{y_{max} - y_T}{x_{max} - x_T} = \frac{1}{\sum_{i \in I} w_i}
 \end{aligned}$$

$$f_{\mathbf{w}}(x) = \begin{cases} x \cdot m_F & \text{if } x_{min} \leq x \leq x_F, \\ y_F + (x - x_F) \cdot m_Z & \text{if } x_F < x < x_T, \\ y_T + (x - x_T) \cdot m_T & \text{if } x_T \leq x \leq x_{max} \end{cases} \quad (95)$$

$$f_{\mathbf{w}}^{-1}(y) = \begin{cases} y/m_F & \text{if } y_{min} \leq y \leq y_F, \\ x_F + (y - y_F)/m_Z & \text{if } y_F < y < y_T, \\ x_T + (y - y_T)/m_T & \text{if } y_T \leq y \leq y_{max} \end{cases} \quad (96)$$

Figure 3 illustrates how a simple linear interpolation between the four boundary points provides interpretability for $\alpha, \beta, \forall i w_i$ and a clear definition of classicality in the domain and range.

Checks for boundary point convergence are also established as parameters fluctuate, ensuring that any division by zero is handled appropriately. This can be especially tricky when working with autograd packages because the existence of such an operation may be allowed — returning *inf* bounds with *NaN* (not a number) gradients and disconnecting the parameter trace.

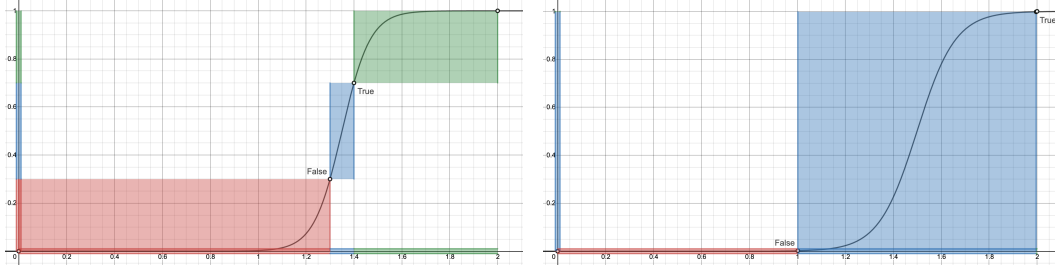


Figure 4: Nonlinear activation function; left: $\alpha = 0.7$, right: maximum α

Figure 4 demonstrates a tailored logistic activation function, a scaled and shifted sigmoid. This is one possible non-linear function that satisfies the True and False boundary requirements with smooth gradients everywhere. The function is defined as:

$$f_{\mathbf{w}}(x) = \frac{1}{1 + e^{(-Ax+B)}} \quad A = \frac{2 \ln((1-\alpha)/\alpha)}{x_F - x_T} \quad B = \ln\left(\frac{\alpha}{1-\alpha}\right) + Ax_F$$

While the asymptotic nature of this activation function may prevent its logical operation from reaching maximally True or False points — $f_{\mathbf{w}}(x) = \{0, 1\}$ — or even allow it due to floating point round-off, the defined regions of classicality still permit the tailored-LNN to operate as usual. It must be noted that this formulation does not satisfy the boundary points under the extrema, $\alpha = 1$. However, no learning would have taken place under this scenario and smooth gradient are therefore not required when operating classically.

By scaling, shifting *and rotating* the logistic function, we can conceivably rectify the intersection between all boundary points for a given α . While the special condition of requiring smoothness without learning, $\alpha = 1$, may be possible by constructing a piecewise-sigmoid, with flat gradients in the classical ranges.

The incorporation of independent linear and non-linear activation functions therefore demonstrates the robustness of the tailored approach and also permits smooth-optimisation techniques to update LNN parameters — future work.

F.7 Proof that zero weighting leads to identity

Intuitively, as a neuron's input weight begins to drop, so too should its influence on the activation function output. An all-but-one zero-weighted neuron should therefore retrieve only the value of the remaining input, represented by an identity activation function.

Here we provide proof from first principles that the design choice of x_F as a function of the maximum weight, w_{max} , is well founded.

Without loss of generality, assume $w_A = 0$, we now show that $w_A A$ & $w_B B = B$ and further visualize it in Figure 5:

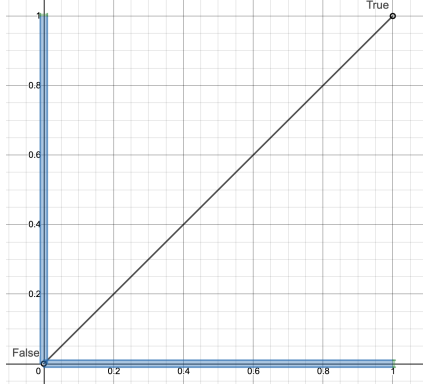


Figure 5: Activation function for a binary conjunction ($\alpha = 1$), with one weight = 0

$$\begin{aligned}
 x_F &= \sum_{i \in I} w_i - \alpha w_{max} = w_B(1 - \alpha) & y_F &= 1 - \alpha \\
 x_T &= \sum_{i \in I} w_i \cdot \alpha = w_B \alpha & y_T &= \alpha \\
 x_{max} &= \sum_{i \in I} w_i = w_B & y_{max} &= 1
 \end{aligned}$$

$$\begin{aligned}
 m_F &= \frac{y_F}{x_F} = \frac{1}{w_B} \\
 m_Z &= \frac{y_T - y_F}{x_T - x_F} = \frac{1}{w_B} \\
 m_T &= \frac{y_{max} - y_T}{x_{max} - y_T} = \frac{1}{w_B}
 \end{aligned}$$

This then leads to:

$$f_{\mathbf{w}}(w_A A + w_B B) = \begin{cases} B & \text{if } x_{min} \leq x \leq x_F, \\ B & \text{if } x_F < x < x_T, \\ B & \text{if } x_T \leq x \leq x_{max} \end{cases} \quad (97)$$

Since the False boundary point is now defined using w_{max} , identity will hold for an n -ary neuron where any number of inputs, but one, are zero-weighted.

Whereas an all-zero weighted input will decrease the identity until it returns an impulse function, at $x = 0$. This results in a True conjunction, False disjunction and a False implication (representative of an ‘if True then False’ statement), making the tailored activation function consistent with its weighted logic counterparts in the upward direction. We note that without directly modifying neuron bounds, the tailored function is incapable of removing contradictions that arise in a dead-state (i.e. all-zero weighted inputs). In this state, it is possible to defer to a learnable β and thereby remove inherent contradictions that may arise without the propagation of bounds — e.g. root disjunction and implication formulae initialised with all rules being True are self-contradictory due to downward inference from a universal quantifier.

F.8 Downward inference

Downward inference may be defined in two ways by weighted fuzzy logic, namely, a functional inverse (Section 4) or logical upward inference (equations (10), (12), (14)). When given the bounds of the output of a neuron as well as the bounds of all the inputs except one, the functional inverse